

ARIN5302 Assignment #3

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1 Part A

1. **A.1:** 6 ; 3 ; 3
2. **A.2:** different; same
3. **A.3:** Query; Key; Value; Query and Key
4. **A.4:** labeled; unlabeled; annotation
5. **A.5:** informative
6. **A.6:** B
7. **A.7:** C
8. **A.8:** B
9. **A.9:**
 - (a) **rigid registration:** rotation + translation; suitable for rigid body alignment like skull and bone.
 - (b) **affine registration:** rotation, translation, scaling, shearing; suitable for global linear alignment.
 - (c) **deformable registration:** non-linear local deformation; suitable for soft tissue alignment like brain or lung.

2 Part B

B.1: (a)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

(b) Given:

$$P = (x, y) = (10, 0), \theta = 30^\circ, t_x = 5, t_y = 3$$

and 2D rigid transformation:

$$\begin{cases} x' = x \cos \theta - y \sin \theta + t_x \\ y' = x \sin \theta + y \cos \theta + t_y \end{cases}$$
$$\begin{aligned} x' &= 10 \times 0.866 - 0 \times 0.5 + 5 \\ &= 8.66 + 5 \\ &= 13.66 \\ y' &= 10 \times 0.5 + 0 \times 0.866 + 3 \\ &= 5 + 3 \\ &= 8 \end{aligned}$$

Final coordinates:

$$(x', y') = (13.66, 8)$$

B.2: (a) Local soft tissue regions.

(b) Deformable registration models local non-linear transformations.

(c) Unreasonable in rigid bone structures due to forced deformation.

B.3: (a) Choose (C): self-supervised pre-training + fine-tuning.

(b) approach (B)-joint training; but approach (C)-two-stage pre-training and fine-tuning.